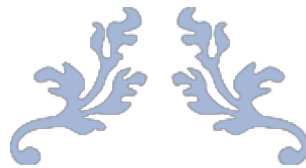


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THE LEONTIEF MODEL: AN APPLICATION OF LINEAR ALGEBRA

Math 2010 Extra Credit Project



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The Leontief Model: An Application of Linear Algebra

By Ayal Yakobe and Dan Jung

Introduction

Introduced in the 1930s, future Nobel Economics laureate Wassily Leontief created one of the most widely used accounting models and their consequent applications remain relevant throughout many economic market scales and fields of study: valid for international, national, and regional scales **and** used today for social accounting models (SAM), general computing equilibriums (CGE), and more.¹² Commonly known as the **Input-Output Model**, or I-O, Leontief's consumption matrix seeks to correlate relationships between various sectors of an economy, tracing the flow of products sold as outputs and commodities bought as inputs. Doing this serves not only to “*bean count*” raw inventory quantities to be stocked within a supply chain, but to correlate their market effect on a particular system of goods being sold to consumers.³

Because of its centrally controlled nature (unlike the “survival of the fittest” purely open market model prevalent in capitalist thinking) Leontief's Input-Output Fixed Proportions Model was heralded by neoclassical supply-side economists. It focused on limiting the “Total Amount Produced of a Particular Good” equal to just the sum of the “Internal Consumer Demand and External Consumer Demand” for such a good. The highly supervised absolute nature of I-O economics really appealed to the centrally planned economies of communist nations, especially

¹ Libretexts. “2.6: Applications – Leontief Models.” *Mathematics LibreTexts*

² Mesnard, Louis de. “Price Consistency in the Leontief Model.”

³ “Input-Output Analysis.” *Corporate Finance Institute*

the former Soviet Union: without any External Market Demand as a purely communist economy, the exact amounts of various goods to be produced for the USSR simply equaled the Production Vector (whose components were just the number of items to be manufactured per type of good) times the Consumption Matrix (whose matrix entries contained the exact fixed proportions of each good, relative to each other, that would ensure no surplus or deficit would exist for any good made or produced in the USSR → resources would be optimally allocated across all State industries → goods would be produced exactly as needed—no more, no less → workers across all State industries would be more likely to be fully employed most of the year.⁴

Interesting!

To avoid food shortages due to commodity crop mismanagement and stay competitive on a manufacturing scale with the United States for defense and space programs, using Leontief's I-O Closed-Form Model made a lot of sense since Communist economies did not believe in International Trade. Hence, there was no External Demand vector in the old USSR. This simplified the mathematics behind the Leontief Model, where $\mathbf{d}$ could safely be set to zero. So, whatever suppliers produced, consumers bought—no unsold goods/resources were to be left unallocated → greater market efficiency across many state-run industries → consistent yields made from agriculture and manufacturing became much more likely for the USSR.

⁴ "Neoclassical Economics." *Corporate Finance Institute*

Theoretical

The model is simple, consisting of an $n \times m$ matrix whose rows represent the production (output) produced by a specific industry. At the same time, the columns denote the allocation (input) of the goods produced in the rows to other distinct sectors. The model can be open or closed – meaning that it should exist in a stable environment that is not apt to change or one that can influence M – and the matrix can be derived using elementary accounting:

$$(x_{i1} + x_{i2} + \cdots + x_{in}) + d_i = X_i$$

Where the **Production Vector X_i** is:

a.) the sum of the **Internal Demand** components $x_{i1} + x_{i2} + \cdots + x_{in}$ (sector by sector)

AND

b.) the **External Demand** for these same goods as d_i (sector by sector).

One can think of **External Demand** as the units of goods that are NOT recycled back into this economic system. The variable a_{ij} can be used to simplify the above equation by assigning it the value of the ratio of a specific good i that is used in the production of another – j . This relationship can be described as:

$$a_{ij} = x_{ij} / X_j$$

where x_{ij} refers to the unique dependency of i required to produce j (demand) and X_j is the total output produced from an industry (supply). By isolating x_{ij} we can rewrite our previous equation as:

$$\sum_{j=1}^n a_{ij} X_j + d_i = X_i$$

where the sum of the proportions required to produce a product given i and j , a_{ij} , is multiplied by the total output that some distinct industry will produce, X_j , and then added with the external demands of another industry, d_i , and equal to its total output, X_i . To emphasize the homeostasis achieved by this equation, certain variables can be swapped, and the model will endure:

$$\sum_{i=1}^n a_{ij} X_i + d_j = X_j$$

Exploring this equation further indicates that a_{ij} can remain a constant, and the final equation is represented simply as the Leontief Input-Output Model in matrix-vector form:

$$A\vec{X} + \vec{d} = \vec{X}$$

Where:

- A is known as the **Consumption Matrix**
- $\vec{X}$ is known as the **Production Vector**
- $A\vec{X}$ is our **Internal Demand Vector**
- $\vec{d}$ is our **External Demand Vector**

In Matrix Form

Sector	Industry 1	Industry 2	...	Industry n	External Demand	Total Output
Industry 1	x_{11}	x_{12}	...	x_{1n}	d_{1n}	X_1 $= \sum_j x_{1j}$
Industry 2	x_{21}	x_{22}	...	x_{2n}	d_{2n}	X_2 $= \sum_j x_{2j}$
...	...	...	...	...	...	...
Industry n	x_{n1}	x_{n2}	...	x_{nn}	d_{nn}	X_n $= \sum_j x_{nj}$

Each column in this table represents a linearly independent vector that makes up the basis of an economic system in $\mathbb{R}^n$ dimensions.⁵

It is important to note that the model requires several assumptions to operate correctly, namely:

- I. All sectors require inputs from each other.
- II. All inputs must be greater than or equal to one.
- III. Inputs are independent of outputs being produced.
- IV. Sectors have one output.
- V. Production of a good from a particular sector will ensure that its output is greater than its input.

⁵ Ravindran, Vish. "Input-Output Model."

- VI. Combinations and scaling of different inputs to produce one unit of output will not be the same.
- VII. There cannot be a negative supply or demand for a good
- VIII. Outputs must be Internal Demand that is contained within a system where $\vec{d} = \vec{0}$, in the case of the **Closed Form** of the Leontief Model.
- IX. Otherwise, if $\vec{d} \neq \vec{0}$, then External Demand will exist and you will have the case of the **Open Form** of the Leontief Model. ⁶



Example

Suppose during Finals Week at Columbia, Dan would like to make some cash, and he decides to sell Red Bull energy drinks, Ramen noodles, and Aleve pain reliever medication at his engineering fraternity house for East Asian STEM grinds like himself, Tau Alpha Pi. He must decide the correct ratio of the items he must purchase as store inventory such that what he buys is guaranteed to sell, given his market-estimated external demand vector representing the top three correlated purchases of Tau Alpha Pi STEM grinds made during previous Finals Weeks. Dan calculates the following matrix A and the external demand **d** vector to be as follows:

$$A = \begin{bmatrix} 0.20 & 0.25 & 0.10 \\ 0.15 & 0.30 & 0.20 \\ 0.10 & 0.15 & 0.25 \end{bmatrix} \quad \vec{d} = \begin{bmatrix} 500 \\ 1100 \\ 1600 \end{bmatrix}$$

The first, second, and third rows of matrix A are the *total outputs* of Red Bull, Ramen, and Aleve, respectively. Likewise, the first, second, and third columns are the *total inputs* of Red

⁶ Laurent, Sofie. "The Mathematical Justification of the Leontief and Sraffa Input-Output Systems."

Bull, Ramen, and Aleve, respectively. How Dan formed his Leontief Open Model Consumption Matrix came from these observations based on previous Finals Week sales at the Tau Alpha Pi Fraternity “Store of Last Resort.” This is what he found:

- For 1 unit item of Red Bull to *sell*, Dan notices it takes:
 - 1.) 0.20 unit items of Red Bull to be bought as inventory → sold at Dan’s store
 - 2.) 0.15 unit items of Ramen to be bought as inventory → sold at Dan’s store
 - 3.) 0.10 unit items of Aleve to be bought as inventory → sold at Dan’s store
- For 1 unit item of Ramen to *sell*, Dan notices that it takes:
 1. 0.25 unit items of Red Bull to be bought as inventory → sold at Dan’s store
 2. 0.30 unit items of Ramen to be bought as inventory → sold at Dan’s store
 3. 0.15 unit items of Aleve to be bought as inventory → sold at Dan’s store
- For 1 unit item of Aleve to *sell*, Dan notices it takes:
 - 1.) 0.10 unit items of Red Bull to be bought as inventory → sold at Dan’s store
 - 2.) 0.20 unit items of Ramen to be bought as inventory → sold at Dan’s store
 - 3.) 0.15 unit items of Aleve to be bought as inventory → sold at Dan’s store
- Restating all three goods, *relative to Red Bull*, as **fixed proportions**, we find that:
 - 1.) To sell x_1 units of Red Bull, Dan must stock $0.20 * x_1$ units of Red Bull
 - 2.) To sell x_2 units of Ramen, Dan must stock $0.25 * x_2$ units of Red Bull
 - 3.) To sell x_3 units of Aleve, Dan must stock $0.10 * x_3$ units of Red Bull

→ Really this

- Restating all three goods, *relative to Ramen*, as **fixed proportions**, we find that:

- 1.) To sell x_1 units of Red Bull, Dan must stock $0.29 * x_1$ units of Ramen
- 2.) To sell x_2 units of Ramen, Dan must stock $0.30 * x_2$ units of Ramen
- 3.) To sell x_3 units of Aleve, Dan must stock $0.10 * x_3$ units of Ramen

- Restating all three goods, *relative to Aleve*, as **fixed proportions**, we find that:

- 1.) To sell x_1 units of Red Bull, Dan must stock $0.20 * x_1$ units of Aleve
- 2.) To sell x_2 units of Ramen, Dan must stock $0.25 * x_2$ units of Aleve
- 3.) To sell x_3 units of Aleve, Dan must stock $0.19 * x_3$ units of Aleve

- Combining these three facts to form a *system of linear equations* yields:

$$0.20x_1 + 0.25 x_2 + 0.10 x_3 = x_1 \quad \text{<Equation 1>}$$

$$0.15x_1 + 0.30 x_2 + 0.20 x_3 = x_2 \quad \text{<Equation 2>}$$

$$0.10x_1 + 0.15 x_2 + 0.25 x_3 = x_3 \quad \text{<Equation 3>}$$

- Converting this system of linear equations to form a 3x3 matrix A will be:

$$A = \begin{bmatrix} 0.20 & 0.25 & 0.10 \\ 0.15 & 0.30 & 0.20 \\ 0.10 & 0.15 & 0.25 \end{bmatrix}$$

Using our Linear Algebra prowess from Math 2010

Okay, so we have the general form of the Leontief Input-Output Model in matrix-vector equation form as:

$$A\vec{X} + \vec{d} = \vec{X}$$

Isolating for the External Demand Vector $\vec{d}$, we get:

$$\vec{d} = \vec{X} - A\vec{X}$$

Factoring out the Production Vector from the RHS of the above equation yields:

$$\vec{d} = (I - A)\vec{X}$$

where **I** is the **Identity Matrix**.

Finally, because we are given a nonzero External Demand Vector $\vec{d}$, we can safely isolate for the Production Vector $\vec{X}$ alone on the LHS of the equation above:

$$\vec{X} = (I - A)^{-1} \vec{d}$$

BUT: We must now calculate the inverse matrix of $[I - A]$, the matrix $[I - A]^{-1}$

To determine the optimal store inventory levels required for Red Bull, Ramen, and Aleve—we subtract matrix A from the identity matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.2 & 0.25 & 0.1 \\ 0.15 & 0.3 & 0.2 \\ 0.1 & 0.15 & 0.25 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.25 & -0.10 \\ -0.15 & 0.70 & -0.2 \\ -0.10 & -0.15 & 0.75 \end{bmatrix}$$

Now we must solve for the determinant of $I - A$ to get its inverse:

$$\begin{aligned}
 \det(I_3 - A) &= \begin{bmatrix} 0.80 & -0.25 & -0.10 \\ -0.15 & 0.70 & -0.20 \\ -0.10 & -0.15 & 0.75 \end{bmatrix} * \begin{bmatrix} 0.80 & -0.25 \\ -0.15 & 0.70 \\ -0.10 & -0.15 \end{bmatrix} \\
 &= [(.80 * .70 * .75) + (-.25 * -.20 * -.10) + (-.10 * -.15 * -.15)] - \\
 &\quad [(-.10 * .70 * -.10) + (-.15 * -.20 * .80) + (.75 * -.15 * -.25)] \\
 &= (.42 + -.005 + -.00225) - (.007 + .024 + .028125) \\
 &= (.41275 - .059125) \\
 &= 0.353625
 \end{aligned}$$

So, the $\det(I - A)$ equals the scalar value of 0.353625.

Now, we need to establish the **Co-factor Matrix [C]**, which is the determinant of each element of the original matrix [A]. This matrix [C], *needed for the calculation of the inverse of the $[I_3 - A]$ matrix*, will be:

$ M_{11} = 0.495 $	$ M_{12} = -0.1325 $	$ M_{13} = 0.0925 $
$ M_{21} = -0.2025 $	$ M_{22} = 0.59 $	$ M_{23} = -0.145 $
$ M_{31} = 0.12 $	$ M_{32} = -0.175 $	$ M_{33} = 0.5225 $

Where:

$$|M_{11}| = |((.7 * .75) - (-.2 * -.15))| = |.495|$$

$$|M_{21}| = |(-.25 * .75) - (-.1 * -.15)| = |-.2025|$$

⋮

$$|M_{33}| = |(.8 * .7) - (-.25 * -.15)| = |.5225|$$

And: **matrix [C] = the Co-factor Matrix**

Our next step on the path to calculate $[I_3 - A]^{-1}$ now will be **to find the transpose of the Co-factor Matrix [C]**, where $[C]^T$ is:

$$[C]^T = \begin{bmatrix} 0.4950 & 0.2025 & 0.1200 \\ 0.1325 & 0.5900 & 0.1750 \\ 0.0925 & 0.1450 & 0.5225 \end{bmatrix}$$

Final Calculation to get the Production Vector $\vec{X}$

Finally, now that we know the *determinant of matrix $[I_3 - A]$* **and matrix $[C]^T$** , we can directly *calculate matrix $[I_3 - A]^{-1}$* , where:

$$[I_3 - A]^{-1} = \frac{1}{\det(I_3 - A)} * [C]^T$$

Numerically, this will yield:

$$[I_3 - A]^{-1} = \left(\frac{1}{0.353625} \right) * \begin{bmatrix} 0.4950 & 0.2025 & 0.1200 \\ 0.1325 & 0.5900 & 0.1750 \\ 0.0925 & 0.1450 & 0.5225 \end{bmatrix}$$

$$= \begin{bmatrix} 1.39979 & 0.572041 & 0.339343 \\ 0.374691 & 1.66843 & 0.494875 \\ 0.261577 & 0.410039 & 1.47755 \end{bmatrix}$$

Recalling that “ $\vec{X} = [(\mathbf{I}_3 - \mathbf{A})]^{-1} * \vec{d}$ ” for our Leontief Open Form Input-Output Market Model, we get:

$$\vec{X} = \begin{bmatrix} 1.39979 & 0.572041 & 0.339343 \\ 0.374691 & 1.66843 & 0.494875 \\ 0.261577 & 0.410039 & 1.47755 \end{bmatrix} * \begin{bmatrix} 500 \\ 1100 \\ 1600 \end{bmatrix}$$

$$= \begin{bmatrix} 1872.75 \\ 2814.42 \\ 2945.92 \end{bmatrix}$$

Finding the primary inputs (**as store inventory to be bought**) required to produce the primary output levels (**as unit items guaranteed to be sold**) of Red Bull, Ramen, and Aleve yield the following: Total Produced = Internal Demand + External Demand, such that:

$$\text{Total Produced: } \vec{X} = \vec{Ax} + \vec{d}$$

$$\text{Internal Demand: } \vec{Ax} = \begin{bmatrix} 0.2 & 0.25 & 0.1 \\ 0.15 & 0.3 & 0.2 \\ 0.1 & 0.15 & 0.25 \end{bmatrix} * \begin{bmatrix} 1872.75 \\ 2814.42 \\ 2945.92 \end{bmatrix} = \begin{bmatrix} 1372.75 \\ 1714.42 \\ 1345.92 \end{bmatrix}$$

$$\text{External Demand: } \begin{bmatrix} 500 \\ 1100 \\ 1600 \end{bmatrix}$$

Now to check our results, let's test our calculated values to see if they equal our given value in our Example, where **External Demand** should equal **Total Produced** minus **Internal Demand**:

$$\text{Verify: Does : } \vec{X} - \vec{Ax} = \vec{d} ?$$

$$\begin{bmatrix} 1872.75 \\ 2814.42 \\ 2945.92 \end{bmatrix} - \begin{bmatrix} 1372.75 \\ 1714.42 \\ 1345.92 \end{bmatrix} = \begin{bmatrix} 500 \\ 1100 \\ 1600 \end{bmatrix}$$

Yes, *exactly*!

Special Case: What if matrix $[I - A]$ is non-invertible?

Two possibilities exist: either no solutions for the Production Vector exist or infinitely many solutions exist for the Production Vector.

- Case 1: No Solutions exist

Suppose Dan found his Leontief Consumption matrix A to be $[A']$, where:

$$A' = \begin{bmatrix} 0.20 & 0.25 & 0.10 \\ 0.15 & 0.30 & 0.20 \\ 0.10 & 0.15 & 1 \end{bmatrix}$$

Since for one unit of Aleve to be sold requires exactly one unit of Aleve to be bought as store inventory, no Aleve is available to help Red Bull and Ramen sell simultaneously, per student purchase at Dan's Store of Last Resort and that would satisfy the external demand for Aleve.

Thus, the Leontief Consumption Model will fail here.



- Case 2: Infinitely many solutions exist

Suppose Dan found his Leontief Consumption matrix A to be $[A'']$, where:

$$A'' = \begin{bmatrix} 0.20 & 0.25 & 0 \\ 0.15 & 0.30 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Here, to sell one unit of Aleve takes exactly one unit of Aleve to be bought as store inventory.

Additionally, no Aleve is needed to help sell Red Bull and Ramen, per student purchase at Dan's

Store of Last Resort. Thus, Dan can choose *any* amount to stock Aleve to be sold because it will not impact the sale of Red Bull and Ramen, per student purchase.



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